Nguyen Viet Hoang Luong

Embedded Systems / Robotics Intern Candidate

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SUMMARY

Control and Automation student focused on embedded firmware, hardware integration, and robotic control.
Built ESP32/PIC prototypes: 3DoF suction arm, PID ball-and-beam, ESP32 remote-controlled robot, and LVGL HMI.

EDUCATION

Hanoi University of Mining and Geology (HUMG) — **Sep 2023 – Expected 2027**
Control Engineering and Automation

AI in Action, VinUniversity — **Feb 28, 2026 – Present**
Applied AI Engineering and project prototyping

PROJECTS

PickPilot – 3DoF Suction Robotic Arm Hardware & Firmware — Jun 2026

Vin AI Project, Team C2-054 – Embedded & Hardware Developer

Tech: ESP32-S3, PlatformIO, NEMA 17, TB6600, 24V vacuum, L298N, WebSocket

- Built hardware and firmware for web-controlled target movement, suction pickup, transfer, and release.
- Integrated ESP32-S3, NEMA 17/TB6600 motion hardware, 24V vacuum, L298N suction driver, and limit switches.
- Mapped web UI target points to the physical robot workspace for repeatable light-object pick-and-place.

Project links: PickPilot | Team C2-054

Ball-and-Beam PID Balancing System — 2026

Personal control-system prototype

Tech: ESP32-S3, MKS SERVO42C, VL53L1X, I2C OLED, PlatformIO, PID, ESP32 Web UI

- Built a 40 cm prototype with ESP32-S3, VL53L1X sensing, MKS SERVO42C actuation, and OLED feedback.
- Tuned PID control to move the ball toward a 20 cm setpoint and hold it near an 18–22 cm target band.
- Debugged oscillation, beam-centering behavior, and long-wire I2C stability on the physical prototype.

ESP32 Remote-Controlled Robot System — 2026

Personal embedded robotics project

Tech: ESP32, ESP32-CAM, ESP-NOW, WebServer, SR04, VL53L0X, IR obstacle sensors, relay control

- Built ESP32 robot control with local web UI, camera preview, and relay-driven water pump.
- Tested SR04, VL53L0X, and IR sensors for distance/obstacle feedback on physical hardware.
- Built an ESP-NOW joystick control link after observing lag in HTTP-based control.

Additional Embedded Projects — 2026

- Built ESP32-C6 IR, AHT30/OLED, LVGL HMI, relay-control, and sensor experiments.
- Built PIC16F887/PIC16F877A boards, PCB prototypes, and nRF24L01 wireless tests.

SKILLS

Firmware: C/C++, Embedded C, PlatformIO, timers, ADC/PWM, UART/I2C/SPI, STEP/DIR.
Motion: PID control, stepper/servo control, motion profiling, sensor feedback, obstacle avoidance.
Interfaces: WebSocket, HTTP Web UI, ESP-NOW, LVGL, IR control, relay control, OLED/I2C.
Hardware: Altium Designer, Proteus, PCB prototyping, hardware integration, actuator/driver selection.
Tools: Git/GitHub, Python, Bash, SolidWorks, AutoCAD.

HONORS

3rd Prize, Leanbot Programming Competition — 2023
Team Leader. Led robot navigation strategy for map traversal, obstacle avoidance, slope movement, and optional line-following using ultrasonic/IR sensors, stepper motors, and A4988 drivers.